COMPLETE DIGITAL ELECTRONICS HANDBOOK

Digital Electronics

A full student textbook for number systems, Boolean algebra, K-Maps, logic families, combinational circuits, flip-flops, registers, counters and semiconductor memories—built from the official model-paper scope and examination pattern.

COURSE CODE
3044

SEMESTER
Third Semester

PROGRAMME
Common to BM / EC / EL

SCHEME
Revision 2021

MAXIMUM MARKS
75

TIME
3 Hours

CREDITS / CATEGORY
Not specified in supplied source

PRIMARY SOURCE
Official Model Question Paper, Sets 1 & 2

Start Learning

FOUR RECONSTRUCTED LEARNING MODULES

From binary digits to complete digital systems

The handbook follows the topic scope and module-outcome codes evidenced in the supplied official-style model paper. Reconstructed titles are clearly identified as handbook organization, not claimed as official syllabus titles.

Module 1
Number systems, gates, Boolean algebra, K-Maps

Module 2

Logic families and combinational circuits

Module 3

Flip-flops, registers, ring and Johnson counters

Module 4

Asynchronous/synchronous counters and memories

COURSE OVERVIEW

What Digital Electronics is and why it matters

Digital electronics represents and processes information using discrete logic levels, normally 0 and 1. It is the foundation of computers, embedded systems, communication equipment, medical instruments, industrial control and modern consumer electronics.

Information as bits

A bit has two states. Groups of bits represent numbers, instructions, characters, addresses and control information.

Logic as decisions

Logic gates implement decisions such as AND, OR, NOT and XOR. Boolean algebra gives a mathematical language to simplify those decisions.

Memory and sequence

Flip-flops and registers remember past information. Counters and state circuits create controlled sequences over time.

Importance and applications

Computing

Processors, memories, storage interfaces and digital buses.

Communication

Encoding, multiplexing, switching, timing and control.

Automation

Programmable controllers, counters, sensors and machine sequencing.

Medical and instrumentation

Digital measurement, data acquisition, alarms and display systems.

Source limitation
The supplied document is a model question paper, not a complete syllabus. Therefore, credits, category, course objectives, exact CO-PO mapping, periods and official reference books are not fabricated. Module titles below are a logical handbook reconstruction from the questions and module-outcome codes.

STUDY METHOD

How to use this handbook

For every topic: understand the idea, inspect the truth table or diagram, work the solved example, then attempt the practice question without looking at the answer.

1. Learn the rule

Read definitions, formulas and design rules.

2. Trace the logic

Follow signal paths, state transitions and timing diagrams.

3. Solve step-by-step

Show all conversions, K-Map groups and design tables.

4. Practise for marks

Use one-mark, three-mark and seven-mark patterns.

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SOURCE AND EXAMINATION INFORMATION

Confirmed examination pattern

The two supplied model sets establish the title, semester, programme applicability, module-outcome codes and paper structure.

Item	Confirmed information
Subject	Digital Electronics
Course code	3044
Semester	Third Semester
Programme	Common to BM / EC / EL
Scheme	Revision 2021
Time	3 hours
Maximum marks	75
Part A	9 questions × 1 mark = 9; answer all
Part B	10 questions; answer any 8; 3 marks each = 24
Part C	6 compulsory groups, each with internal OR; 7 marks per group = 42
Available module-outcome codes	M1.01, M1.03, M1.04, M2.02, M2.04, M3.01, M3.02, M3.03, M3.04, M4.01, M4.02, M4.03, M4.04
Not specified in supplied source	Credits, category, periods, formal objectives, CO-PO mapping, official reference books

LEARNING ROADMAP

Four reconstructed modules

The following module titles are a handbook organization inferred from the topic distribution in the model papers.

Module	Evidenced outcome codes	Core topics	Exam emphasis
1. Number Systems, Logic Gates, Boolean Algebra and K-Maps	M1.01, M1.03, M1.04	Conversions, complements, binary arithmetic, universal gates, Boolean laws, De Morgan, K-Maps	Short calculations and 7-mark conversion / minimization problems
2. Logic Families and Combinational Logic Circuits	M2.02, M2.04	CMOS/ECL parameters, adders, multiplexers, demultiplexers, binary-to-Gray conversion	3-mark features and 7-mark truth-table/K-Map designs
3. Flip-Flops, Shift Registers, Ring and Johnson Counters	M3.01, M3.02, M3.03, M3.04	Sequential logic, flip-flops, conversions, register types, ring and Johnson sequence	Symbols, truth tables, diagrams and conversion/design questions
4. Counters and Semiconductor Memories	M4.01, M4.02, M4.03, M4.04	Ripple/synchronous counters, MOD design, timing diagrams, RAM/ROM/PROM/EPROM/EEPROM	Counter design and memory comparisons

MODULE 1 • M1.01 / M1.03 / M1.04

Number Systems, Logic Gates, Boolean Algebra and K-Maps

This title is reconstructed from the official model-paper topics. The chapter develops the complete calculation and logic-design foundation required by those questions.

Learning targets

- Convert between decimal, binary and hexadecimal—including fractions.
- Perform binary arithmetic and complement subtraction.
- Use logic gates and universal-gate implementation.
- Simplify expressions using Boolean laws and K-Maps.

Core exam skills

- Show each conversion step.
- Use the specified bit width for complements.
- Draw symbols and truth tables neatly.
- Mark K-Map groups and write final reduced expression.

1. Number systems and positional value

A number system is defined by its radix or base. In a positional number system, the value of a digit depends on both the digit and its position.

(d_n...d_1d_0.d_{-1}d_{-2})_r = Σ d_i r^i

System	Base	Digits	Example
Decimal	10	0–9	$(125.4)_{10}$
Binary	2	0,1	$(10101.01)_2$
Hexadecimal	16	0–9, A–F	$(4BAC)_{16}$

Place-value chart

Position	...	3	2	1	0	-1	-2
Weight in base r	...	r^3	r^2	r^1	r^0	r^{-1}	r^{-2}

Hexadecimal digit table

Hex	A	B	C	D	E	F
Decimal	10	11	12	13	14	15
Binary	1010	1011	1100	1101	1110	1111

Malayalam Support
Radix □□□□□□ base □□□. Binary-□ □□□ position-□□□ 2□□□□ power; hexadecimal-□ 16□□□□ power □□□□□□□□□□□□□□.

2. Conversion methods

Binary → Decimal

Multiply each bit by its place value and add.

$$(10101)_2 = 1\times2^4+0\times2^3+1\times2^2+0\times2^1+1\times2^0 = 21$$

Decimal integer → Binary

Repeatedly divide by 2 and read remainders upward.

$$35_{10} = 100011_2$$

Decimal fraction → Binary

Repeatedly multiply the fraction by 2 and read integer parts downward.

$$0.625_{10} = .101_2$$

Hex → Decimal

Multiply each hexadecimal digit by powers of 16.

$$(AB6)_{16} = 10\times16^2+11\times16+6 = 2742_{10}$$

Hex ↔ Binary

Replace every hexadecimal digit by four binary bits.

$(4BAC)_{16} = 0100\ 1011\ 1010\ 1100_2$

Decimal → Hex

Repeatedly divide integer part by 16. Use A–F for remainders 10–15.

$125_{10} = 7D_{16}$

Interactive binary counter

Use the buttons to observe binary states and decimal value.

0 0 0 0

Decimal: 0

3. Binary arithmetic

Operation	Rules
Addition	0+0=0; 0+1=1; 1+0=1; 1+1=10; 1+1+1=11
Subtraction	0−0=0; 1−0=1; 1−1=0; 0−1 requires borrow
Multiplication	0×anything=0; 1×bit=bit
Division	Same long-division idea as decimal, but quotient digits are only 0 or 1

Binary addition example

```
101101
+ 011011
--------
1001000
```

45 + 27 = 72, so the result is verified.

Binary multiplication example

```
1011
× 101
-------
1011
0000
101100
```

```
-------
110111
```

$11 \times 5 = 55 = 110111_2$.

4. Signed numbers and complements

One's complement is formed by inverting every bit. Two's complement is one's complement plus 1. In an n-bit signed two's-complement system, the leftmost bit is the sign bit.

$$\text{One's complement: } 0 \leftrightarrow 1$$

$$\text{Two's complement} = \text{One's complement} + 1$$

Example: 8-bit two's complement of 14

1. 14 = 00001110
2. Invert bits → 11110001
3. Add 1 → 11110010

Subtraction: 46 − 14

```
46 = 00101110
14 = 00001110
2's comp(14) = 11110010

00101110
+ 11110010
-----------
1 00100000
```

Discard the end carry. Result = $00100000_2 = 32_{10}$.

Malayalam Support
Two's complement subtraction-□ subtract □□□□□□□□□□ number-□□□□ bits invert □□□□□□□ 1 □□□□□□□□. □□□□□ minuend-□□□□ add □□□□□□□□. End carry □□□□□□□□□□□ □□□ discard □□□□□□□□.

Common mistake
Use the required bit width before complementing. Do not complement an unpadded number.

5. Logic levels and gates

In positive logic, the higher voltage level represents logic 1 and the lower voltage level represents logic 0. A logic gate performs a Boolean operation on one or more binary inputs.

Logic-gate symbol guide

AND is D-shaped, OR has a curved input, NOT is triangular, and inversion is shown by a small circle.

AND

$$Y = AB$$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

OR

$$Y = A + B$$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

NOT

$$Y = \overline{A}$$

A	Y
0	1
1	0

NAND

$Y = \overline{(AB)}$

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

NOR

$Y = \overline{(A+B)}$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

XOR

$Y = A \oplus B$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Gate classification

Basic: AND, OR, NOT. **Universal:** NAND, NOR. **Special:** XOR, XNOR.

6. Universal gates

A universal gate can implement any Boolean function. NAND and NOR are universal because NOT, AND and OR can be created using only one gate type.

Function using NAND	Connection	Result
NOT	Join both NAND inputs to A	A NAND A = $\overline{A}$
AND	NAND A,B, then invert output with NAND	$[\overline{(AB)}] = AB$
OR	Invert A and B using NAND, then NAND the results	$\overline{(\overline{A}\,\overline{B})} = A+B$
XOR	Four-NAND standard network	$A \oplus B$

Malayalam Support
Universal gate □□□□□□□ □□□□□□□□□□ basic gates-□□ □□□□□□□□□□□□□□ □□□□□□□□□□ gate □□□. NAND □□□□□□□ □□□□□□□□□□□□ NOR □□□□□□□ □□□□□□□□□□□□ □□□□□□□ digital circuit □□□□□□□□□□□□□.

7. Boolean algebra

Law	Expression
Identity	$A+0=A$; $A\cdot 1=A$
Null	$A+1=1$; $A\cdot 0=0$
Idempotent	$A+A=A$; $A\cdot A=A$
Complement	$A+\overline{A}=1$; $A\cdot\overline{A}=0$
Involution	$\overline{(\overline{A})}=A$
Commutative	$A+B=B+A$; $AB=BA$
Associative	$A+(B+C)=(A+B)+C$; $A(BC)=(AB)C$
Distributive	$A(B+C)=AB+AC$; $A+BC=(A+B)(A+C)$
Absorption	$A+AB=A$; $A(A+B)=A$
De Morgan 1	$\overline{(A+B)}=\overline{A}\,\overline{B}$
De Morgan 2	$\overline{(AB)}=\overline{A}+\overline{B}$

Algebraic simplification example

$$
\begin{aligned}
Y &= \overline{A}B + AB + A\overline{B} \\
&= B(\overline{A} + A) + A\overline{B} \\
&= B + A\overline{B} \\
&= (B + A)(B + \overline{B}) \\
&= A + B
\end{aligned}
$$

8. SOP, POS, minterms and maxterms

A minterm is an AND term containing every variable once. A maxterm is an OR term containing every variable once. Canonical SOP is a sum of minterms; canonical POS is a product of maxterms.

$$F(A,B,C) = \Sigma m(1,3,5,7)$$

$$F(A,B,C) = \Pi M(0,2,4,6)$$

9. Karnaugh Maps

A K-Map arranges adjacent cells in Gray-code order so that one variable changes between neighbouring cells. Groups must contain 1, 2, 4, 8 or 16 cells.

Two-variable K-Map

A\B	0	1
0	m0	m1
1	m2	m3

Three-variable K-Map

A\BC	00	01	11	10
0	m0	m1	m3	m2
1	m4	m5	m7	m6

Four-variable K-Map

AB\CD	00	01	11	10
00	m0	m1	m3	m2
01	m4	m5	m7	m6
11	m12	m13	m15	m14
10	m8	m9	m11	m10

Grouping rules

- Use largest possible groups.
- Every required 1 must be covered.
- Overlapping is allowed.
- Wrap-around adjacency is allowed.
- Diagonal cells are not adjacent.
- Don’t-care cells may be used only when helpful.

Typical reduction workflow

1. Place all minterms and don’t-cares.
2. Form essential largest groups.
3. Write one product term per group.
4. OR the terms and verify coverage.

Solved K-Map: $F(A,B,C)=\Sigma m(0,2,3,4,5,6)$

Place 1s at m0,m2,m3,m4,m5,m6.

Group m0,m2,m4,m6 as a quad → $\overline{C}$.

Group m4,m5 as a pair → $A\overline{B}$.

Group m2,m3 as a pair → $\overline{A}B$.

Final: $F = \overline{C} + A\overline{B} + \overline{A}B$.

Four-variable example with don’t-cares

$F(A,B,C,D)=\Sigma m(1,4,7,10,13)+\Sigma d(5,14,15)$.

Use d-cells only to enlarge groups:

m1,m5(d),m13,m? cannot form a valid full quad unless every required cell is adjacent.

One possible minimal cover is found by mapping and selecting valid pairs/quads.

Always show the marked map, groups, reduced terms and final SOP. Do not treat every don't-care as mandatory 1.

Module 1 expected exam questions

One mark

- Find decimal value of 10101_2.
- Write 1’s or 2’s complement.
- Name a universal gate.

Three marks

- Convert a hexadecimal integer/fraction.
- State De Morgan’s theorems.
- Reduce a small expression using K-Map.

Seven marks

- Mixed number-system operations.
- Universal-gate implementation.
- Four-variable K-Map with don’t-cares.

MODULE 2 • M2.02 / M2.04

Logic Families and Combinational Logic Circuits

This reconstructed chapter covers the logic-family parameters and combinational design problems evidenced in the official model paper.

Learning targets

- Compare TTL, CMOS and ECL.
- Explain fan-out, delay and power dissipation.
- Design adders, multiplexers and demultiplexers.
- Derive a four-bit binary-to-Gray converter.

Combinational principle

Output depends only on present inputs. There is no stored state and normally no clock.

> Combinational circuit-□□□□ output □□□□□□□□□□□ input-□□ □□□□□□□ □□□□□□□□□□□□□□□□; previous state □□□□□□□□□□.

1. Logic families and parameters

A logic family is a group of compatible digital ICs built using a common circuit technology and logic-level convention.

Parameter	Meaning
Fan-in	Maximum number of inputs a gate is designed to accept
Fan-out	Number of standard gate inputs one output can drive reliably
Propagation delay	Time between input transition and corresponding output transition
Power dissipation	Average power consumed by a gate
Noise margin	Maximum unwanted noise voltage tolerated without logic error
Speed-power product	Propagation delay × power dissipation; lower is generally better

Feature	TTL	CMOS	ECL
Device technology	Bipolar transistor	MOSFET	Bipolar differential switching
Power dissipation	Moderate	Very low in static condition	High
Speed	Moderate / fast	Modern CMOS can be very fast	Traditionally fastest
Noise immunity	Moderate	High	Moderate
Input impedance	Moderate	Very high	Lower than CMOS
Model-paper emphasis	General comparison	Least power dissipation; CMOS features	Fastest family; ECL features

Malayalam Support
Propagation delay □□□□□□□ input change □□□□□□□□□□ □□□□ output change □□□□□□ □□□□□□□□□□□ □□□□□□ □□□□□□□□. Fan-out □□□ output □□□□ gate inputs drive □□□□□□□□□□□□□ □□□□□□□□□.

2. Combinational vs sequential circuits

Feature	Combinational	Sequential
Output depends on	Present inputs only	Present inputs and previous state
Memory	No	Yes
Clock	Normally not required	Usually required
Examples	Adder, MUX, DEMUX, converter	Flip-flop, register, counter

3. Half adder and full adder

Half adder

$$\text{Sum } S = A \oplus B$$

$$\text{Carry } C = AB$$

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Full adder

$$S = A \oplus B \oplus C_{in}$$

$$C_{out} = AB + AC_{in} + BC_{in}$$

A	B	Cin	S	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Full adder using two half adders

HA1 adds A and B. HA2 adds the first sum and Cin. OR the two carry outputs.

4. Parallel adder

A multi-bit addition requires one full adder per bit. The carry output of one stage feeds the carry input of the next stage. A four-bit parallel adder adds $A_3A_2A_1A_0$ and $B_3B_2B_1B_0$ simultaneously, while carry ripples from LSB to MSB.

Diagram guidance

Draw the corresponding block or logic diagram with standard labels, signal direction and input/output names. Use the truth table and equations beside this panel to verify the drawing.

5. Multiplexers

A multiplexer is a data selector. It selects one of many input lines and connects it to one output according to select lines.

For 2^n inputs, number of select lines = n

2 inputs → 1 select **4 inputs → 2 selects** **8 inputs → 3 selects** **16 inputs → 4 selects**

Diagram guidance

Draw the corresponding block or logic diagram with standard labels, signal direction and input/output names. Use the truth table and equations beside this panel to verify the drawing.

4:1 MUX: $Y = I_0\overline{S}_1\overline{S}_0 + I_1\overline{S}_1S_0 + I_2S_1\overline{S}_0 + I_3S_1S_0$

S1	S0	Selected input
0	0	I0
0	1	I1
1	0	I2
1	1	I3

Interactive 4:1 MUX

Set input bits and select lines. The selected input appears at Y.

S1S0: 00 **Y:** 0

Malayalam Support
Multiplexer □□ input lines-□ □□□□□□□ select lines □□□□□□□□□□□ □□□ input □□□□□□□ output-□□□□□□□ □□□□□□□□□□□□□□□□□□ data selector □□□.

6. Demultiplexers

A demultiplexer is a data distributor. One input is routed to one of several outputs according to select lines.

S1	S0	Active output in 1:4 DEMUX
0	0	Y0 = D
0	1	Y1 = D
1	0	Y2 = D
1	1	Y3 = D

$Y_0=D\ \overline{S}_1\overline{S}_0;\ Y_1=D\ \overline{S}_1S_0;\ Y_2=D\ S_1\overline{S}_0;\ Y_3=D\ S_1S_0$

7. Binary-to-Gray converter

Gray code changes only one bit between adjacent numbers. This reduces transition ambiguity in encoders and position systems.

MSB: $G_3 = B_3$

Remaining: $G_2=B_3 \oplus B_2$; $G_1=B_2 \oplus B_1$; $G_0=B_1 \oplus B_0$

Binary B3B2B1B0	Gray G3G2G1G0
0000	0000
0001	0001
0010	0011
0011	0010
0100	0110
0101	0111
0110	0101
0111	0100
1000	1100
1001	1101
1010	1111
1011	1110
1100	1010
1101	1011
1110	1001
1111	1000

Design method

1. Prepare Binary-to-Gray conversion table.
2. Write each Gray output column as a function of B3,B2,B1,B0.
3. Plot each output on a K-Map.
4. Reduced results:
   G3=B3
   G2=B3⊕B2
   G1=B2⊕B1
   G0=B1⊕B0
5. Implement using one direct connection and three XOR gates.

Module 2 expected exam questions

One mark

- Fastest logic family.
- Least-power logic family.
- Select lines for 8:1 MUX.

Three marks

- Features of CMOS/ECL.
- Define fan-out and propagation delay.
- Draw D/T or half-adder table.

Seven marks

- Design full adder.
- Explain 1:4 DEMUX.
- Design 4-bit Binary-to-Gray converter.

MODULE 3 • M3.01 / M3.02 / M3.03 / M3.04

Flip-Flops, Shift Registers, Ring and Johnson Counters

This reconstructed chapter develops the sequential-logic topics repeatedly tested in the official model paper.

Learning targets

- Distinguish combinational and sequential logic.
- Use SR, JK, D and T flip-flops.
- Convert JK to T and JK to D.
- Trace register, ring and Johnson sequences.

Sequential principle

Output depends on present input and stored state. A clock coordinates state changes.

> Sequential circuit previous state □□□□□□□□□□. Clock edge □□□□□□□□□ state update □□□□□□□.

1. Clock and state concepts

A clock is a periodic digital waveform. Edge-triggered flip-flops respond at a rising edge or falling edge. Present state is the current stored output; next state is the value after the active clock edge.

Diagram guidance

Draw the corresponding block or logic diagram with standard labels, signal direction and input/output names. Use the truth table and equations beside this panel to verify the drawing.

2. Flip-flop summary

Flip-flop	Inputs	Key action	Important condition
SR	S,R	Set, reset or hold	S=R=1 invalid for active-high NOR form
JK	J,K	Improved SR; toggle when J=K=1	Avoid race-around with proper triggering
D	D	Next Q follows D	Data storage / delay
T	T	Toggle when T=1	Counters and divide-by-2

SR truth table

S	R	Q(n+1)	Meaning
0	0	Qn	Hold
0	1	0	Reset
1	0	1	Set
1	1	Invalid	Forbidden

JK truth table

J	K	Q(n+1)
0	0	Qn
0	1	0
1	0	1
1	1	$\overline{Q}$n

D truth table

D	Q(n+1)
0	0
1	1

$$Q(n+1)=D$$

T truth table

T	Q(n+1)
0	Qn
1	$\overline{Q}$n

$$Q(n+1)=T\oplus Qn$$

3. SR flip-flop using NAND gates

A cross-coupled NAND latch normally uses active-low inputs $\bar{S}$ and $\bar{R}$. Both inputs high hold the state; pulling $\bar{S}$ low sets Q; pulling $\bar{R}$ low resets Q. Both low is invalid.

Malayalam Support
NAND SR latch-□ inputs active-low □□□. $\bar{S}$=0 set, $\bar{R}$=0 reset. □□□□□□ 0 □□□□□□□□□.

4. Flip-flop conversion

Conversion means forcing an available flip-flop to behave like a desired flip-flop. Use the desired next state and the available flip-flop excitation table, then simplify input equations.

JK → T

For T=0, hold. For T=1, toggle. A JK flip-flop already toggles for J=K=1.

$$J = T, K = T$$

JK → D

To make Q(n+1)=D:

$$J = D, K = \bar{D}$$

Malayalam Support
Flip-flop conversion-□ desired flip-flop-□□□□ next state □□□□□ □□□□□□. Available flip-flop □ state □□□□□□□□□□ J,K □□□□□□□ inputs □□□□□□□□□ □□□□□ excitation table □□□□□□□□□□ □□□□□□□□□□.

5. Shift registers

A shift register is a chain of flip-flops that stores and moves binary data one position per clock pulse.

Diagram guidance

Draw the corresponding block or logic diagram with standard labels, signal direction and input/output names. Use the truth table and equations beside this panel to verify the drawing.

Type	Input	Output	Use
SISO	Serial	Serial	Delay line
SIPO	Serial	Parallel	Serial-to-parallel conversion
PISO	Parallel	Serial	Parallel-to-serial conversion
PIPO	Parallel	Parallel	Temporary parallel storage

Interactive four-bit shift register

Choose the next serial input and press Clock.

0 0 0 0

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□□□ clock pulse-□□□ data □□□ flip-flop-□ □□□□□□ □□□□□□ flip-flop-□□□□□□ shift □□□□□□□.
SISO serial input/serial output; SIPO serial input/parallel output.

6. Ring counter

A ring counter circulates a single 1 (or single 0) through n flip-flops. It normally provides n valid states and requires initialization.

$$n \text{ flip-flops} \rightarrow n \text{ ring-counter states}$$

Clock	Q3Q2Q1Q0
Initial	1000
1	0100
2	0010
3	0001
4	1000

7. Johnson counter

A Johnson counter feeds the complemented output of the last flip-flop back to the first. An n-stage Johnson counter produces 2n valid states.

$$n \text{ flip-flops} \rightarrow 2n \text{ Johnson states}$$

Clock	Q3Q2Q1Q0
0	0000
1	1000
2	1100
3	1110
4	1111
5	0111
6	0011
7	0001
8	0000

Ring vs Johnson
Ring uses direct feedback and normally n states. Johnson uses complemented feedback and produces 2n states.

Module 3 expected exam questions

One mark

- Which circuits require clock input?
- Name SISO register.
- State Johnson states for n stages.

Three marks

- Draw D and T truth tables.
- Applications of shift registers.
- Draw 4-bit Johnson counter.

Seven marks

- JK-to-T and JK-to-D conversion.
- Explain PISO register.
- Explain ring or Johnson counter.

MODULE 4 • M4.01 / M4.02 / M4.03 / M4.04

Counters and Semiconductor Memories

This reconstructed chapter covers asynchronous and synchronous counter design, timing and memory types tested in the official model paper.

Learning targets

- Explain ripple and synchronous counters.
- Design MOD-6, MOD-8 and MOD-10 counters.
- Trace up/down timing diagrams.
- Compare RAM, ROM, PROM, EPROM and EEPROM.

Counter definition

A counter is a sequential circuit that advances through a prescribed state sequence in response to clock pulses.

1. MOD number and flip-flop requirement

Choose smallest n such that $2^n \geq N$

$n = \text{ceiling}(\log_2 N)$

Counter	Required flip-flops
MOD-6	3
MOD-8	3
MOD-10	4
MOD-16	4

2. Asynchronous and synchronous counters

Feature	Asynchronous (ripple)	Synchronous
Clocking	Only first FF receives external clock; others are clocked by preceding output	All FFs receive common clock
Output change	Ripples stage by stage	Changes nearly simultaneously
Speed	Lower due to cumulative delay	Higher
Hardware	Simpler	More combinational control logic
Glitches	More likely during transitions	Reduced with proper design

Diagram guidance

Draw the corresponding block or logic diagram with standard labels, signal direction and input/output names. Use the truth table and equations beside this panel to verify the drawing.

Ripple frequency division: $f_{out} = f_{in}/2^n$

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Ripple counter-□ clock change □□□ flip-flop □□□□□□ □□□□ □□□□□□□□□. Synchronous counter-□ □□□□□ flip-flops-□□□ □□□ clock □□□□□□□□□□□□□□□ state □□□□□□□ □□□□□.

3. Three-bit ripple up and down counters

A three-bit ripple up counter counts 000,001,010,011,100,101,110,111. A ripple down counter follows the reverse sequence. Whether Q or $\bar{Q}$ clocks the next stage depends on flip-flop triggering convention.

Decimal	Up Q2Q1Q0	Down Q2Q1Q0
0	000	111
1	001	110
2	010	101
3	011	100
4	100	011
5	101	010
6	110	001
7	111	000

4. Truncated asynchronous counters

A MOD-N counter with N not equal to a power of two uses reset logic to force the counter back to 000… after the Nth state.

MOD-6 design

1. $N=6$; choose 3 flip-flops because $2^3=8\geq6$.
2. Valid states: 000 through 101.
3. Detect 110 using reset/decode logic.
4. Clear all flip-flops to 000.

MOD-10 design

1. $N=10$; choose 4 flip-flops.
2. Valid states: 0000 through 1001.
3. Detect 1010.
4. Apply asynchronous clear to return to 0000.

5. Synchronous counter design

All flip-flops share the clock. The design procedure uses a state table, flip-flop excitation requirements, K-Maps and simplified input equations.

1. List present and next states.
2. Choose flip-flop type.
3. Use excitation table to find required inputs.
4. Minimize each input using K-Map.
5. Draw common-clock circuit and verify sequence.

> 3-bit synchronous up counter using T FFs: $T_0=1$; $T_1=Q_0$; $T_2=Q_1Q_0$

> 3-bit synchronous down counter using T FFs: $T_0=1$; $T_1=\bar{Q}_0$; $T_2=\bar{Q}_1\bar{Q}_0$

Interactive 3-bit counter

0	0	0

Decimal state: 0

6. Semiconductor memories

A semiconductor memory stores binary words at numbered locations. Address lines select a location; data lines transfer a word.

$$\text{Memory capacity} = \text{Number of locations} \times \text{bits per location}$$

Memory	Volatile?	Write method	Typical use
RAM	Yes	Electrical, repeated	Temporary working data
SRAM	Yes	Flip-flop cells; no refresh	Cache / fast buffers
DRAM	Yes	Capacitor cells; refresh required	Main memory
ROM	No	Fixed or programmed content	Permanent firmware
PROM	No	Programmed once	One-time configuration
EPROM	No	UV erasure, then reprogram	Older development systems
EEPROM	No	Electrical erase/rewrite	Settings, calibration, firmware

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RAM power off □□□□□□□□ data □□□□□□□□□□□□; working data-□□□□□□□. ROM family power off □□□□□□□□□□ data □□□□□□□□□□□□. PROM once, EPROM UV erase, EEPROM electrically erase □□□□□□□□.

7. RAM vs ROM

Point	RAM	ROM
Nature	Read/write memory	Primarily read memory
Volatility	Volatile	Non-volatile
Speed	Generally faster for working storage	Depends on technology
Purpose	Temporary data and programs in execution	Permanent instructions / firmware

Module 4 expected exam questions

One mark

- Name the synchronous counter.
- Flip-flops for MOD-10.
- Memory used for working data.

Three marks

- Asynchronous vs synchronous.
- Types of RAM.
- RAM vs ROM.

Seven marks

- Three-bit ripple up/down counter.
- MOD-6 or MOD-10 design.
- Synchronous up/down counter design.

DIGITAL LOGIC BANKS

Rules, formulas, symbols and design summaries

Use these compact banks for rapid revision and as a checklist while solving design problems.

Formula and rule bank

1's complement: invert every bit

2's complement: 1's complement + 1

MUX: 2^n inputs require n select lines

Counter FF count: smallest n with $2^n \geq N$

Ripple division: $f_{out}=f_{in}/2^n$

> Half adder: $S=A\oplus B$; $C=AB$

> Full adder: $S=A\oplus B\oplus C_{in}$; $C_{out}=AB+AC_{in}+BC_{in}$

> Johnson states=2n; Ring states=n

Binary-to-Gray bank

> $G_n=B_n$

> $G_i=B_{i+1}\oplus B_i$

Copy the MSB. XOR each adjacent pair of binary bits for the remaining Gray bits.

K-Map rule bank

- Groups: 1,2,4,8,16 cells.
- Use largest groups.
- Cover every required 1.
- Overlap if useful.
- Wrap-around allowed.
- No diagonal grouping.
- Don’t-cares are optional.

Memory capacity bank

> Capacity = locations × bits per location

Example: 1K×8 = 1024 locations × 8 bits = 8192 bits = 1024 bytes.

Number-system conversion bank

From	To	Method
Binary	Decimal	Weighted sum of powers of 2
Decimal integer	Binary	Repeated division by 2
Decimal fraction	Binary	Repeated multiplication by 2
Hex	Binary	Four bits per hex digit
Binary	Hex	Group bits in fours from radix point
Decimal	Hex	Repeated division / multiplication by 16

Logic-gate symbol and truth-table bank

Diagram guidance

Draw the corresponding block or logic diagram with standard labels, signal direction and input/output names. Use the truth table and equations beside this panel to verify the drawing.

Gate	Boolean expression	Output 1 condition
AND	$Y=AB$	All inputs 1
OR	$Y=A+B$	At least one input 1
NOT	$Y=\overline{A}$	Input 0
NAND	$Y=\overline{(AB)}$	Except all inputs 1
NOR	$Y=\overline{(A+B)}$	All inputs 0
XOR	$Y=A\oplus B$	Inputs different
XNOR	$Y=\overline{(A\oplus B)}$	Inputs equal

Combinational circuit bank

Circuit	Inputs / Outputs	Core equation or idea
Half adder	A,B → S,C	$S=A\oplus B$; $C=AB$
Full adder	A,B,Cin → S,Cout	Two half adders + OR
4:1 MUX	I0–I3, S1,S0 → Y	One selected input
1:4 DEMUX	D,S1,S0 → Y0–Y3	One selected output
Binary-to-Gray	B3–B0 → G3–G0	MSB direct, adjacent XORs

Flip-flop and sequential bank

Device	Hold	Set	Reset	Toggle / follow
SR	S=0,R=0	1,0	0,1	1,1 invalid
JK	0,0	1,0	0,1	1,1 toggles
D	—	D=1	D=0	Qnext=D
T	T=0	—	—	T=1 toggles

Timing diagram bank

When drawing timing diagrams, align vertical transition lines with the correct active clock edge. In a ripple counter, show delayed changes from Q0 to Q1 to Q2. In a synchronous counter, show all output changes referenced to the same clock edge.

Diagram guidance

Draw the corresponding block or logic diagram with standard labels, signal direction and input/output names. Use the truth table and equations beside this panel to verify the drawing.

Counter design bank

Task	Essential steps
Asynchronous MOD-N	Choose FF count; list valid states; detect first invalid state; reset
Synchronous counter	State table; excitation table; K-Maps; input equations; common-clock circuit
Up/down tracing	Write state sequence and timing; note triggering convention
Frequency division	Each binary stage divides frequency by 2

Memory comparison bank

Type	Erase / program method	Retains data without power?
RAM	Electrical read/write repeatedly	No
ROM	Factory or fixed program	Yes
PROM	One-time electrical programming	Yes
EPROM	UV erase, electrical program	Yes
EEPROM	Electrical erase and rewrite	Yes

SOLVED EXAMPLES

Exam-style worked problems

Each solution shows the method and final answer rather than only a hint.

1. Convert 125_{10} to hexadecimal

125 ÷ 16 = 7 remainder 13 (D)

Read quotient and remainder: 7D

Answer: $(125)_{10} = (7D)_{16}$

2. Convert $(124.56)_{16}$ to decimal

Integer part: $1\times16^2 + 2\times16 + 4 = 256+32+4 = 292$

Fraction: $5\times16^{-1} + 6\times16^{-2} = 5/16 + 6/256 = 0.3125 + 0.0234375 = 0.3359375$

Answer: 292.3359375_{10}

3. Binary addition: 35 + 19

$35_{10} = 100011_2$

$19_{10} = 010011_2$

```
  100011
+ 010011
--------
  110110
```

Answer: $54_{10} = 110110_2$

4. Eight-bit two's-complement subtraction: 46−14

46 = 00101110

14 = 00001110

1's comp of 14 = 11110001

2's comp = 11110010

00101110 + 11110010 = 1 00100000

Discard carry.

Answer: $00100000_2 = 32_{10}$

5. Universal-gate conversion

NOT: A NAND A = $\overline{A}$

AND: NAND A,B gives $\overline{(AB)}$; NAND that output with itself gives AB.

OR: NAND A,A gives $\overline{A}$; NAND B,B gives $\overline{B}$; NAND the two gives A+B by De Morgan.

Thus NAND alone can implement the basic gates.

6. Half-adder design

Truth table gives S=1 for 01 and 10, so $S=A\oplus B$.

Carry is 1 only for 11, so C=AB.

Implement one XOR gate for Sum and one AND gate for Carry.

7. Select lines for an 8:1 MUX

For 2^n inputs, select lines=n.

$8=2^3$, therefore n=3.

Answer: 3 select lines.

8. Four-bit binary to Gray

Input B3B2B1B0=1011

G3=B3=1

G2=B3⊕B2=1⊕0=1

G1=B2⊕B1=0⊕1=1

G0=B1⊕B0=1⊕1=0

Answer: 1110 Gray

9. JK to T conversion

T=0 requires hold; JK hold is J=K=0.

T=1 requires toggle; JK toggle is J=K=1.

Therefore J=T and K=T.

10. Four-bit Johnson sequence

Start 0000. Feed complement of Q3 to Q0.

0000→1000→1100→1110→1111→0111→0011→0001→0000.

A four-stage Johnson counter has 2n=8 states.

11. Flip-flops for MOD-10

Find smallest n such that $2^n \geq 10$.

$2^3=8<10$; $2^4=16\geq10$.

Answer: 4 flip-flops.

12. Memory capacity

A 2K×8 memory has 2048 locations and 8 bits/location.

Capacity=2048×8=16384 bits=2048 bytes=2 KB.

MODULE-WISE EXPECTED QUESTIONS

High-priority practice patterns

These are practice patterns inferred from the supplied model papers; they are not guaranteed examination questions.

Module 1

Frequently tested

- Mixed conversion and complement problem.
- Universal gates and NAND implementation.
- Boolean laws or De Morgan.
- 3/4-variable K-Map with don't-cares.

Module 2

Important design

- CMOS/ECL comparison and parameters.
- Half/full adder design.
- 4:1 MUX or 1:4 DEMUX.
- Binary-to-Gray converter.

Module 3

Important diagram

- Flip-flop truth/function tables.
- JK-to-T or JK-to-D conversion.
- Shift-register operation.
- Ring/Johnson diagram and sequence.

Module 4

Important timing

- Ripple up/down counter with timing.
- MOD-6/MOD-10 asynchronous design.
- Synchronous up/down design.
- RAM/ROM/PROM/EPROM/EEPROM comparison.

PRACTICE SECTION

Module-wise exercises and self-assessment

Attempt without looking at the answer key. Final answers are supplied below.

Module 1 practice

1. Convert 93_{10} to binary.
2. Convert $3A7_{16}$ to decimal.
3. Find 8-bit two's complement of 00110101.
4. Subtract 27 from 52 using 8-bit two's complement.

5. Simplify $F(A,B,C)=\Sigma m(1,3,5,7)$.

Module 2 practice

1. How many select lines for 16:1 MUX?
2. Write full-adder equations.
3. Design 2:1 MUX equation.
4. Convert binary 1101 to Gray.
5. Give three CMOS features.

Module 3 practice

1. Complete JK table.
2. Show JK-to-D equations.
3. Trace 1011 through a four-stage SISO register.
4. List four shift-register applications.
5. Write 4-bit ring and Johnson sequences.

Module 4 practice

1. Flip-flops required for MOD-12.
2. Write 3-bit up sequence.
3. State MOD-6 reset state.
4. Compare SRAM and DRAM.
5. Differentiate PROM, EPROM and EEPROM.

Objective practice

No.	Question	Options
1	Logic family with traditionally least static power	A TTL • B CMOS • C ECL • D RTL
2	An 8:1 MUX requires	A 2 • B 3 • C 4 • D 8 select lines
3	Output depends on present input and past state in	A combinational • B sequential • C decoder • D adder
4	Four-stage Johnson counter has	A 4 • B 6 • C 8 • D 16 states
5	Temporary working data is usually stored in	A ROM • B RAM • C PROM • D EPROM

Practice answer key

Module 1: 1) 1011101_2. 2) 935_{10}. 3) 11001011. 4) $25_{10} = 00011001_2$. 5) F=C.

Module 2: 1) 4. 2) S=A⊕B⊕Cin; Cout=AB+ACin+BCin. 3) $Y=I0\overline{S}+I1S$. 4) 1011. 5) Very low static power, high input impedance, good noise immunity.

Module 3: JK: hold/reset/set/toggle; JK-to-D J=D,$K=\overline{D}$; remaining answers follow chapter tables.

Module 4: 4 FFs; 000→001→010→011→100→101→110→111; reset when 110 appears; SRAM faster/no refresh, DRAM denser/refresh; PROM once, EPROM UV, EEPROM electrical.

Objective: 1-B, 2-B, 3-B, 4-C, 5-B.

Self-assessment checklist

- ✓ I can show full number-conversion steps.
- ✓ I can draw all basic and universal gate symbols.
- ✓ I can group a four-variable K-Map correctly.
- ✓ I can derive adder, MUX and converter equations.
- ✓ I can trace flip-flop, register and counter states.
- ✓ I can compare memory types without mixing volatility and programming method.

QUICK REVISION

Last-day revision sheet

Use this section after completing the detailed chapters.

Numbers

- Binary weights: powers of 2.
- Hex digit = 4 bits.
- 2’s comp = invert + 1.

Logic

- NAND/NOR universal.
- De Morgan swaps AND/OR and complements terms.
- XOR=inputs different.

K-Maps

- Gray order 00,01,11,10.
- Largest power-of-two groups.
- Wrap yes; diagonal no.

Combinational

- Half/full adder equations.
- 2^n MUX inputs need n selects.
- Gray uses adjacent XORs.

Flip-flops

- SR set/reset.
- JK 11 toggles.
- D follows data.
- T 1 toggles.

Registers

- SISO/SIPO/PISO/PIPO.
- One shift per clock.
- Ring=n states; Johnson=2n.

Counters

- Async ripple; sync common clock.
- Choose n with $2^n \geq N$.
- Each stage divides by 2.

Memory

- RAM volatile.
- ROM non-volatile.
- PROM once; EPROM UV; EEPROM electrical.

Common mistakes

Wrong bit width in complements; binary/hex grouping from wrong side; non-Gray K-Map ordering; diagonal grouping; missing carry term in full adder; confusing MUX with DEMUX; failing to initialize ring counter; wrong reset state in MOD counter; calling ROM volatile.

FRESH MODEL QUESTION PAPER

DIGITAL ELECTRONICS • Course 3044 • 3 Hours • 75 Marks

Fresh questions are based on the confirmed official structure and topic distribution; they are not copied from the supplied sets.

PART A

Answer all questions in one word, one sentence or a small symbol. Each carries 1 mark. (9×1=9)

1. Write the binary equivalent of decimal 23.
2. Write the two's complement of 01011010.
3. Name the gate whose output is 1 when its two inputs are different.
4. Which logic family is traditionally noted for very high speed?
5. How many select lines are required for a 16:1 multiplexer?
6. Name the flip-flop whose next output directly follows its data input.
7. How many valid states are produced by a 5-stage Johnson counter?
8. How many flip-flops are required for a MOD-12 counter?
9. Name the non-volatile memory that can be erased electrically.

PART B

Answer any eight. Each carries 3 marks. (8×3=24)

1. Convert $(3B.8)_{16}$ to decimal.
2. State both De Morgan's theorems.
3. Implement NOT and AND using NAND gates only.
4. Define fan-out, propagation delay and power dissipation.
5. Write the truth table and expressions of a half adder.
6. Write the output equation and function table of a 2:1 multiplexer.
7. Write the truth tables of D and T flip-flops and one application of each.
8. List four applications of shift registers.
9. Give three differences between asynchronous and synchronous counters.
10. Compare SRAM and DRAM.

PART C

Answer one alternative from every group. Each group carries 7 marks. (6×7=42)

1. **Module 1:** (A) Perform: (i) 157_{10} to hexadecimal, (ii) 61−25 using 8-bit two's complement, (iii) $7D3_{16}$ to binary. **OR** (B) Simplify F(A,B,C,D)=Σm(0,2,5,7,8,10,13,15) using a K-Map.
2. **Module 2:** (A) Design a full adder from its truth table and realize it using two half adders. **OR** (B) Explain a 1:4 demultiplexer with function table, equations, logic diagram description and applications.
3. **Module 3:** (A) Convert a JK flip-flop into D and T flip-flops using conversion tables and equations. **OR** (B) Explain a four-bit Johnson counter with diagram description and state sequence.
4. **Module 3:** (A) Explain PISO and SIPO shift registers with operation tables. **OR** (B) Explain an SR flip-flop using NAND gates, including active-low truth table and invalid condition.
5. **Module 4:** (A) Explain a three-bit ripple up counter with logic and timing diagrams. **OR** (B) Compare RAM and ROM and explain PROM, EPROM and EEPROM.
6. **Module 4 design:** (A) Design a MOD-6 asynchronous counter using T flip-flops. **OR** (B) Design a three-bit synchronous down counter using T flip-flops.

COMPLETE MODEL ANSWERS

Full answers for every fresh-model question

Both alternatives in every Part C group are answered so students can study the complete pattern.

Part A exact answers

No.	Answer
1	10111_2
2	10100110
3	XOR gate
4	ECL
5	4 select lines
6	D flip-flop
7	10 states
8	4 flip-flops
9	EEPROM

Part B answers

B1. Convert $(3B.8)_{16}$ to decimal

$3\times16^1 + 11\times16^0 + 8\times16^{-1} = 48 + 11 + 0.5 = 59.5$

Answer: $(3B.8)_{16} = 59.5_{10}$

B2. De Morgan's theorems

First: $\overline{(A+B)} = \overline{A}\,\overline{B}$

Second: $\overline{(AB)} = \overline{A} + \overline{B}$

They are used to move complements across AND/OR operations while interchanging the operation.

B3. NOT and AND using NAND

NOT: connect both NAND inputs to A, so Y=A NAND A=$\overline{A}$.

AND: first X=$\overline{(AB)}$ using a NAND. Then Y=X NAND X=$\overline{X}$=AB.

B4. Logic-family parameters

Fan-out: number of standard gate inputs driven by one output.

Propagation delay: time between input transition and output response.

Power dissipation: average electrical power consumed by a gate, commonly measured in mW.

B5. Half adder

Truth table: 00→S0,C0; 01→1,0; 10→1,0; 11→0,1.

Sum $S=A\oplus B$; Carry $C=AB$.

Use XOR for Sum and AND for Carry.

B6. 2:1 multiplexer
Function: S=0 selects I0; S=1 selects I1.
Equation: $Y=I0\overline{S}+I1S$.
The select line enables one AND path; an OR gate combines the paths.

B7. D and T flip-flops
D: D=0→Qnext=0; D=1→Qnext=1. Application: data register.
T: T=0→hold; T=1→toggle. Application: binary counter / frequency divider.

B8. Shift-register applications
Serial-to-parallel conversion, parallel-to-serial conversion, temporary data storage, digital delay line, sequence generation and data transfer. Any four earn full credit.

B9. Asynchronous vs synchronous counters
Asynchronous: only first FF gets external clock; output ripples; cumulative delay makes it slower.
Synchronous: all FFs share common clock; outputs change together; faster but needs more control logic.

B10. SRAM vs DRAM
SRAM uses bistable cells, is fast, needs no refresh and has lower density/higher cost.
DRAM uses capacitor cells, needs refresh, has higher density/lower cost and is used as main memory.

Part C full answers

C1A. Mixed number-system operations
i) 157_{10} to hexadecimal:
157 ÷ 16 = 9 remainder 13(D)
Answer: $9D_{16}$.

ii) 61−25 using 8 bits:
61 = 00111101
25 = 00011001
1’s complement of 25 = 11100110
2’s complement = 11100111

```
  00111101
+ 11100111
-----------
1 00100100
```

Discard carry. $00100100_2 = 36_{10}$.

iii) $7D3_{16}$ to binary:
7=0111, D=1101, 3=0011
Answer: $0111\ 1101\ 0011_2$.

C1B. K-Map simplification

$F(A,B,C,D)=\Sigma m(0,2,5,7,8,10,13,15)$.

Place 1s on the four-variable K-Map in Gray order.

Group m0,m2,m8,m10 as a quad. In this group B=0 and D=0, so term $\overline{B}\overline{D}$.

Group m5,m7,m13,m15 as a quad. Here B=1 and D=1, so term BD.

Final simplified expression:

$F = \overline{B}\overline{D} + BD = B \text{ XNOR } D$.

A and C disappear because they change inside each quad.

C2A. Full-adder design

Inputs: A,B,Cin. Outputs: S,Cout.

Truth table:

000→00, 001→10, 010→10, 011→01,

100→10, 101→01, 110→01, 111→11.

From K-Map:

$S=A\oplus B\oplus Cin$.

$Cout=AB+ACin+BCin$.

Realization using two half adders:

HA1: $X=A\oplus B$, $C1=AB$.

HA2: $S=X\oplus Cin$, $C2=X\cdot Cin$.

$Cout=C1+C2$.

This gives a complete full adder.

C2B. 1:4 demultiplexer

A 1:4 DEMUX routes data input D to one of four outputs.

Function table:

S1S0=00→Y0=D

01→Y1=D

10→Y2=D

11→Y3=D

Equations:

$Y0=D\ \overline{S}1\ \overline{S}0$

$Y1=D\ \overline{S}1\ S0$

$Y2=D\ S1\ \overline{S}0$

$Y3=D\ S1\ S0$

Logic: two NOT gates generate complements; four 3-input AND gates produce outputs.

Applications: data routing, serial distribution, communication switching and memory selection.

C3A. JK conversion to D and T

JK excitation:
$0 \to 0$ requires J=0,K=X
$0 \to 1$ requires J=1,K=X
$1 \to 0$ requires J=X,K=1
$1 \to 1$ requires J=X,K=0

$JK \to D$:
Desired Qnext=D.
K-Map/excitation reduction gives J=D and $K=\overline{D}$.
Thus Qnext follows D.

$JK \to T$:
T=0 requires hold, so J=K=0.
T=1 requires toggle, so J=K=1.
Therefore J=T and K=T.
Draw one inverter for K in $JK \to D$ and direct T connections in $JK \to T$.

C3B. Four-bit Johnson counter

Use four D flip-flops with common clock. Connect $Q0 \to D1$, $Q1 \to D2$, $Q2 \to D3$ and feed $\overline{Q}3$ back to D0.
Starting at 0000:
$0000 \to 1000 \to 1100 \to 1110 \to 1111 \to 0111 \to 0011 \to 0001 \to 0000$.
There are 2n=8 valid states.
Because complemented feedback is used, it is called a twisted-ring counter.
Applications: timing sequence generation, divide-by-8 sequence and non-overlapping control signals.

C4A. PISO and SIPO registers

SIPO:
Data enters serially into first flip-flop. Each clock shifts bits one stage. After four clocks, Q3,Q2,Q1,Q0 provide the parallel word.
Example serial input 1,0,1,1 from reset:
Clock1 1000; Clock2 0100; Clock3 1010; Clock4 1101 (orientation depends on drawing).

PISO:
Parallel inputs are loaded simultaneously using LOAD control. In SHIFT mode, each clock transfers one stored bit to serial output.
Applications: communication interfaces and reducing wire count.
Diagrams should show four flip-flops, common clock, serial path and load/shift selection.

C4B. SR latch using NAND

The cross-coupled NAND latch has active-low $\overline{S}$ and $\overline{R}$.

$\overline{S}$ $\overline{R}$:

1 1 → hold

0 1 → set Q=1

1 0 → reset Q=0

0 0 → invalid because both outputs become 1 and the final state after release is uncertain.

Two NAND outputs are cross-coupled, giving memory by positive feedback.

Never apply both active-low inputs at 0 simultaneously.

C5A. Three-bit ripple up counter

Use three T flip-flops with T=1. External clock drives FF0. Q0 clocks FF1 and Q1 clocks FF2 according to the chosen edge convention.

State sequence:

000,001,010,011,100,101,110,111, then 000.

Q0 frequency=fin/2; Q1=fin/4; Q2=fin/8.

In timing diagram Q0 toggles every clock, Q1 every two clocks and Q2 every four clocks.

Because each stage responds after the previous one, transitions ripple and cumulative delay occurs.

C5B. RAM, ROM, PROM, EPROM and EEPROM

RAM is volatile read/write memory for temporary working data.

ROM is non-volatile memory used for fixed programs or firmware.

PROM is programmed once by the user.

EPROM can be erased using ultraviolet light and reprogrammed.

EEPROM can be erased and rewritten electrically.

RAM is used during active computation; ROM-family devices retain information without power.

C6A. MOD-6 asynchronous counter

Required modulus N=6.

Flip-flops: smallest n with $2^n \geq 6$ is n=3.

Use three T flip-flops with T=1 as a ripple counter.

Valid states: 000,001,010,011,100,101.

The next natural state is 110. Decode Q2=1 and Q1=1 (state 110) using a NAND/AND reset network according to active-clear polarity.

Apply asynchronous clear to all flip-flops, forcing 000.

Unused states 110 and 111 are eliminated by reset. Draw three T FFs, ripple clock connections and reset decoder.

C6B. Three-bit synchronous down counter using T flip-flops

All T flip-flops receive the same clock.

Down sequence: 111,110,101,100,011,010,001,000, then 111.

T input requirements:

$T0=1$ because Q0 toggles every clock.

$T1=\overline{Q}0$ because Q1 toggles when lower bit Q0 is 0.

$T2=\overline{Q}1\overline{Q}0$ because Q2 toggles when both lower bits are 0.

Connect these equations using NOT and AND gates. Since all flip-flops share the clock, state changes are synchronous.

REFERENCE LEARNING LINKS

Supplementary resources

The supplied source does not specify official textbooks. These links are supplementary and are not claimed as official course references.

Official model-paper page

SITTTR model question-paper page for course 3044

NPTEL

NPTEL course portal — search for Digital Circuits / Digital Electronics.

All About Circuits

for supplementary reading.

FINAL EXAMINATION CHECKLIST

Before you submit the answer book

Calculations

- ✓ Base and bit width shown.
- ✓ Complement steps shown.
- ✓ Carry treatment shown.

- ✓ Final answer labelled with base.

Design answers

- ✓ Truth/state table included.
- ✓ K-Map groups marked.
- ✓ Reduced equations written.
- ✓ Diagram labelled and operation explained.

Final exam tip

For a seven-mark design question, a bare final circuit is not enough. Include the table, reduction, equations, circuit and working explanation.